CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 13: Statistics

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the class mark and the class size of the class 35–55.

2. Classes 0–10, 10–20, 20–30, 30–40, 40–50 have frequencies 7, 8, 12, 13, 10. Find the mean by the direct method.

3. Classes 100–120, 120–140, 140–160, 160–180, 180–200 have frequencies 12, 14, 8, 6, 10. Find the mean by the step-deviation method.

4. Classes 0–20, 20–40, 40–60, 60–80, 80–100 have frequencies 6, 8, 10, 12, 4. Find the modal class and the mode.

5. Classes 0–10, 10–20, 20–30, 30–40, 40–50 have frequencies 8, 16, 36, 34, 6. Find the median.

6. For the data in the previous question, find the mode.

7. The mean of a distribution is 30 and its median is 32. Use the empirical relationship to estimate the mode.

8. Classes 0–10, 10–20, 20–30, 30–40, 40–50 have frequencies 4, 6, p, 6, 4 and the mean is 25. Explain why p cannot be found from this information alone.

9. Classes 0–20, 20–40, 40–60, 60–80, 80–100 have frequencies 5, f, 10, 8, 7, and the mean is 50. Find f.

10. Convert the classes 10–19, 20–29, 30–39 into continuous classes.

Answer Key

1. Class mark 45; class size 20
2. $\frac{1360}{50} = 27.2$
3. $a = 150$: $150 + \frac{-12}{50} \times 20 = 145.2$
4. Modal class 60–80; mode $= 60 + \frac{2}{10} \times 20 = 64$
5. $n = 100$; median class 20–30; median $= 20 + \frac{50 - 24}{36} \times 10 \approx 27.22$
6. $20 + \frac{36 - 16}{72 - 16 - 34} \times 10 \approx 29.09$
7. $3 \times 32 - 2 \times 30 = 36$
8. The data is symmetric about 25, so the mean is 25 for every value of p.
9. $\frac{1740 + 30f}{30 + f} = 50$ gives $f = 12$
10. 9.5–19.5, 19.5–29.5, 29.5–39.5

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